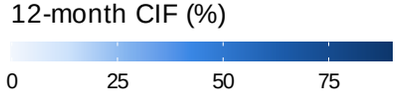
Supplementary: unadjusted 12-month cumulative incidence (Aalen–Johansen)

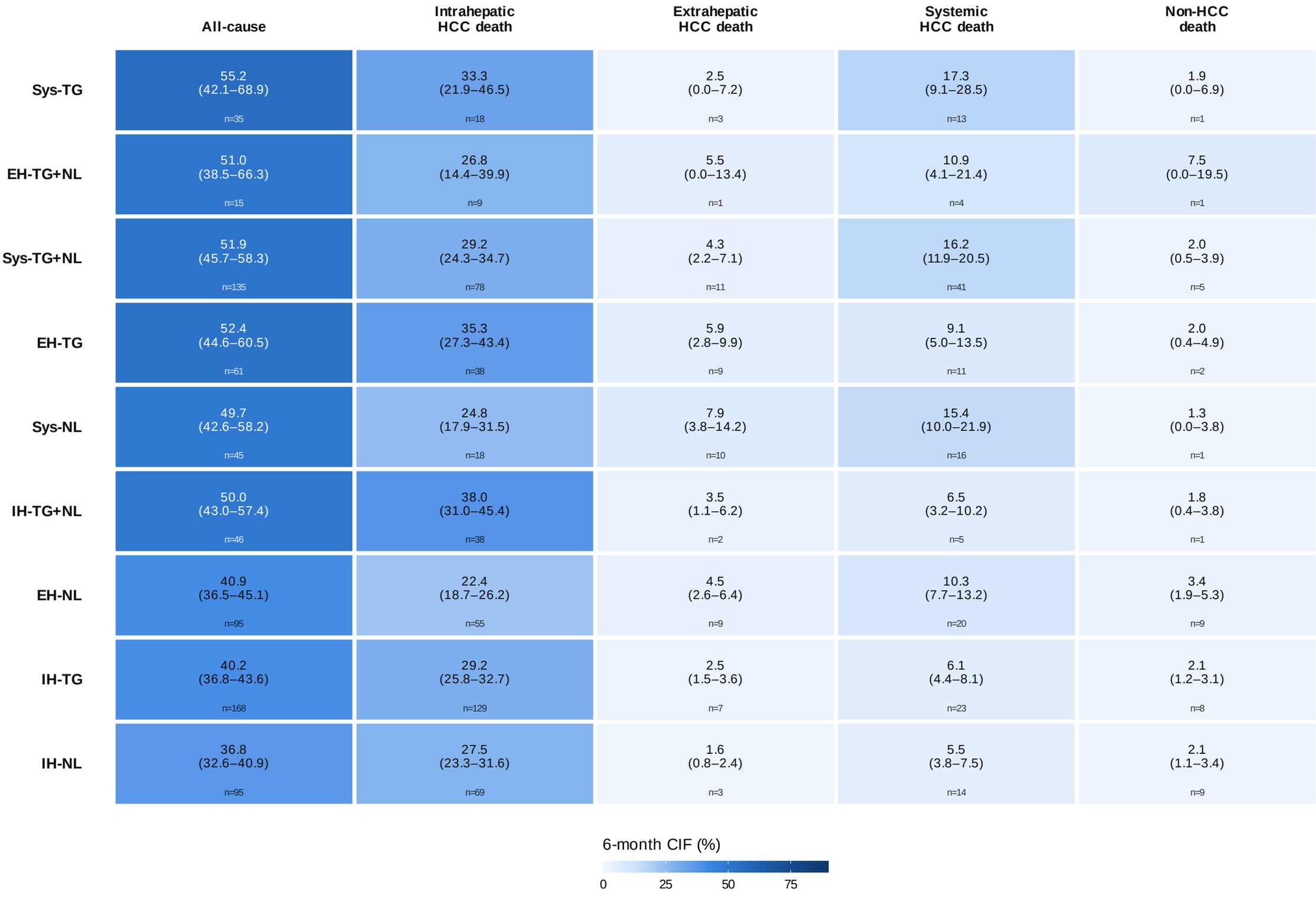
% (95% CI); all-cause = 1 − Kaplan–Meier; n = events within 12 months

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
Sys-TG	78.7 (62.3–88.0) n=36	39.0 (27.1–56.1) n=18	6.3 (2.1–18.9) n=3	31.1 (20.1–48.1) n=14	2.4 (0.3–16.3) n=1
EH-TG+NL	74.0 (51.2–86.2) n=21	35.0 (21.2–57.8) n=10	7.2 (1.9–27.3) n=2	20.9 (10.3–42.8) n=6	10.9 (3.7–31.7) n=3
Sys-TG+NL	79.5 (72.7–84.6) n=154	45.4 (38.9–53.1) n=88	6.8 (4.0–11.4) n=13	24.2 (18.8–31.1) n=47	3.1 (1.4–6.9) n=6
EH-TG	63.4 (53.1–71.4) n=72	36.4 (28.6–46.4) n=42	10.0 (5.7–17.5) n=11	13.5 (8.4–21.6) n=15	2.7 (0.9–8.1) n=3
Sys-NL	81.0 (69.2–88.2) n=63	33.0 (24.1–45.4) n=26	16.8 (10.2–27.7) n=13	28.5 (19.9–40.6) n=22	2.7 (0.7–10.6) n=2
IH-TG+NL	73.0 (62.3–80.7) n=74	53.8 (44.7–64.6) n=55	6.2 (2.9–13.6) n=6	9.9 (5.5–17.8) n=10	3.2 (1.0–9.7) n=3
EH-NL	62.9 (56.2–68.6) n=159	29.3 (24.2–35.5) n=76	7.3 (4.7–11.4) n=18	19.7 (15.3–25.5) n=48	5.9 (3.6–9.6) n=15
IH-TG	66.8 (61.7–71.3) n=269	47.4 (42.7–52.6) n=193	5.0 (3.2–7.7) n=19	9.6 (7.1–13.1) n=38	4.0 (2.5–6.5) n=16
IH-NL	49.8 (43.9–55.1) n=161	33.4 (28.6–39.0) n=109	2.6 (1.3–5.1) n=8	9.2 (6.5–13.0) n=29	4.7 (2.8–7.7) n=15



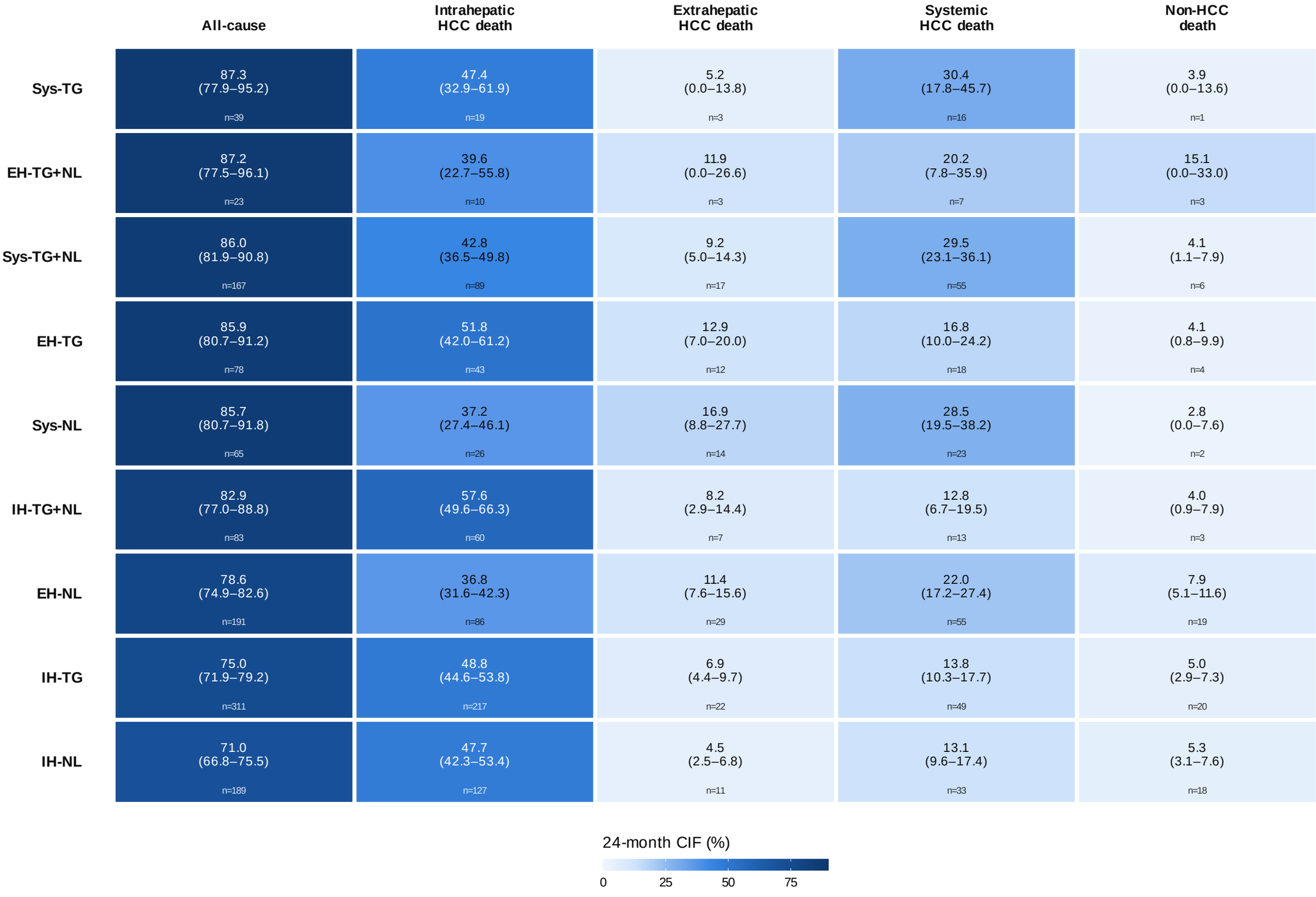
Supplementary: adjusted 6-month CIF after first progression

Same models as Figure 4.4A; n = events within 6 months



Supplementary: adjusted 24-month CIF after first progression

Same models as Figure 4.4A; n = events within 24 months



A2 Absolute risk difference at 6 months vs population-standardised average

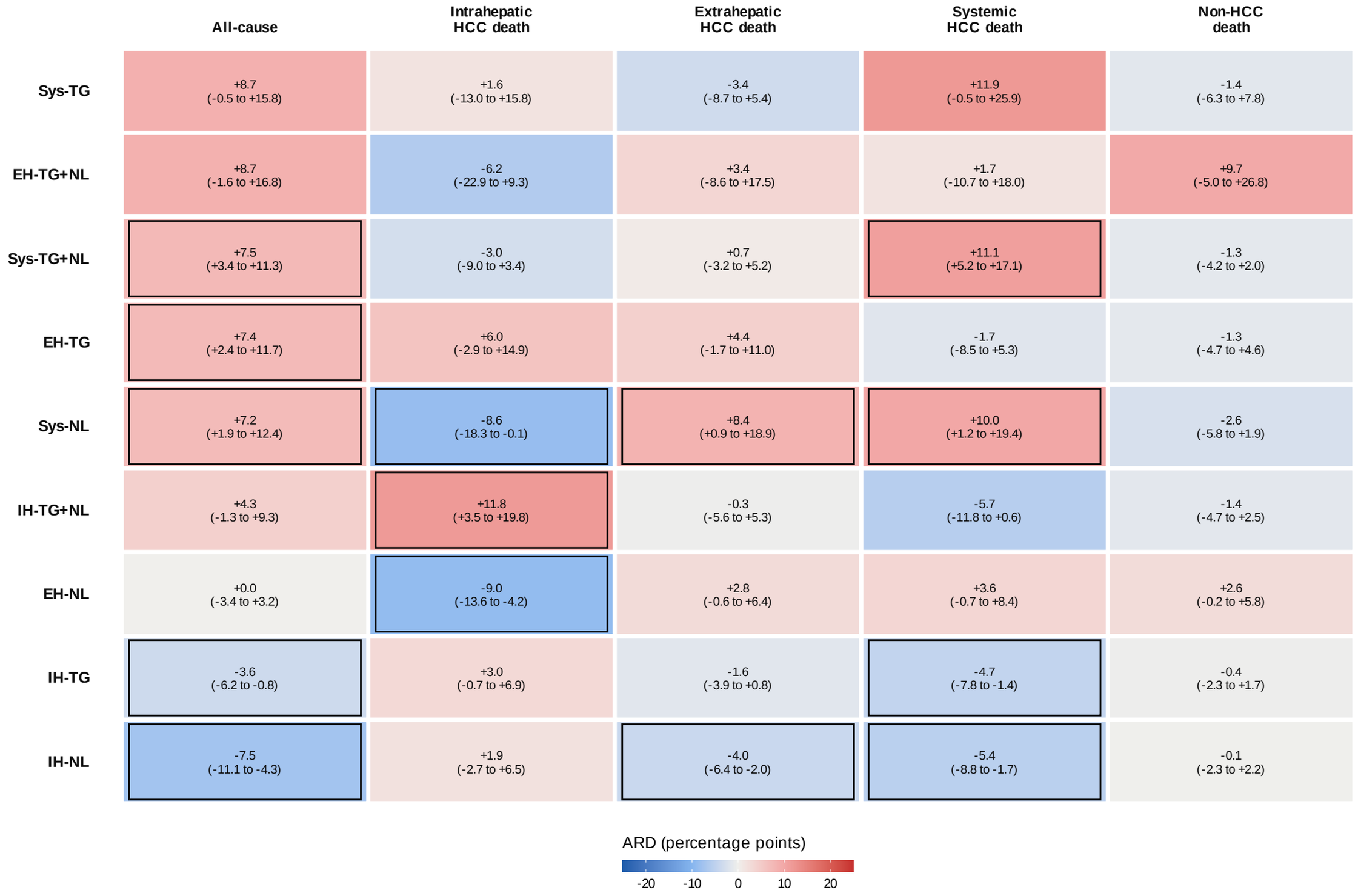
ARD = standardised CIF of the pattern − weighted average over patterns (weights = observed pattern shares); boxed: 95% CI excludes 0



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

A2 Absolute risk difference at 24 months vs population-standardised average

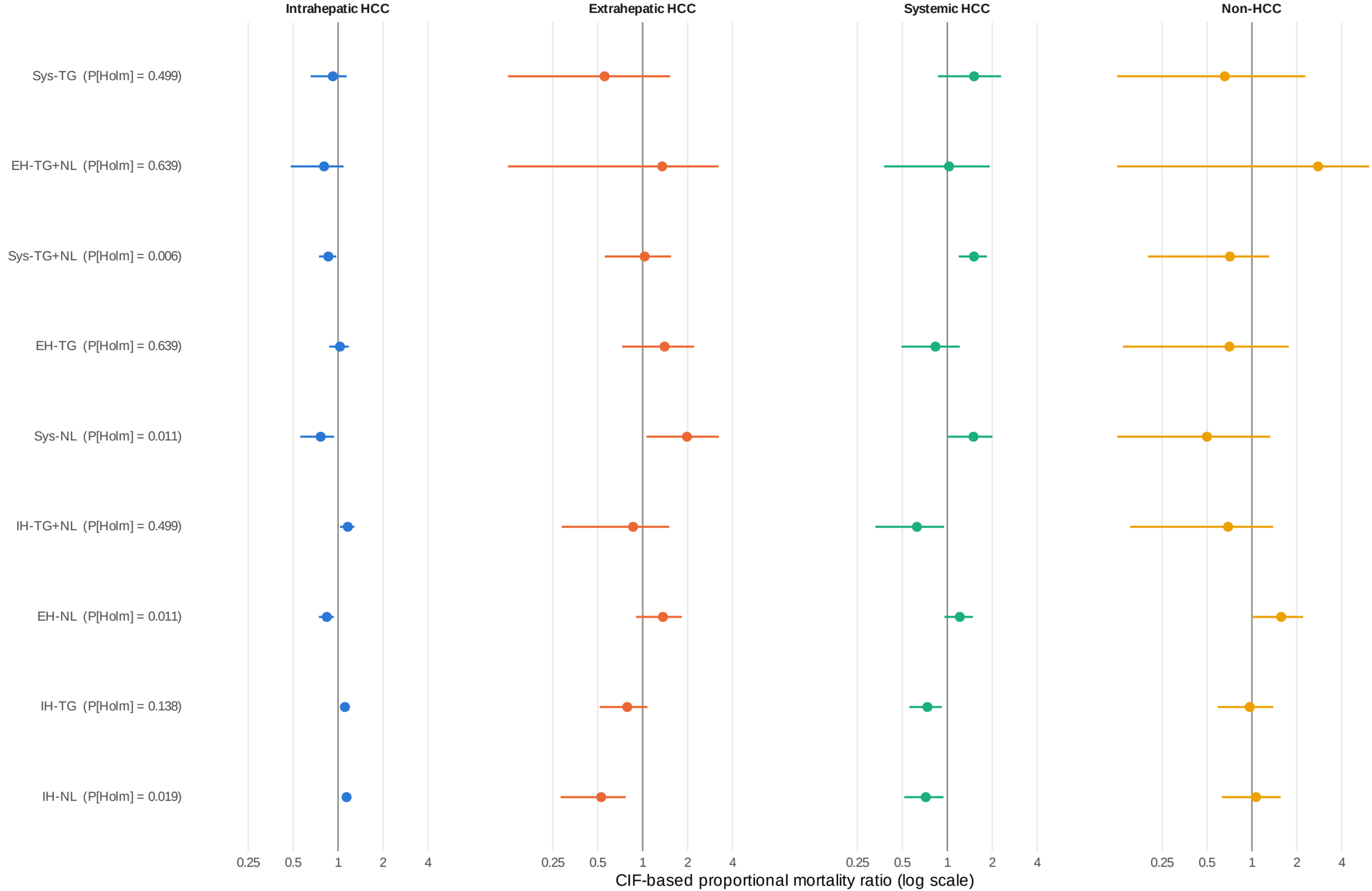
ARD = standardised CIF of the pattern − weighted average over patterns (weights = observed pattern shares); boxed: 95% CI excludes 0



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

A4 Cause-of-death enrichment: CIF-based proportional mortality ratio (PMR) at 6 months

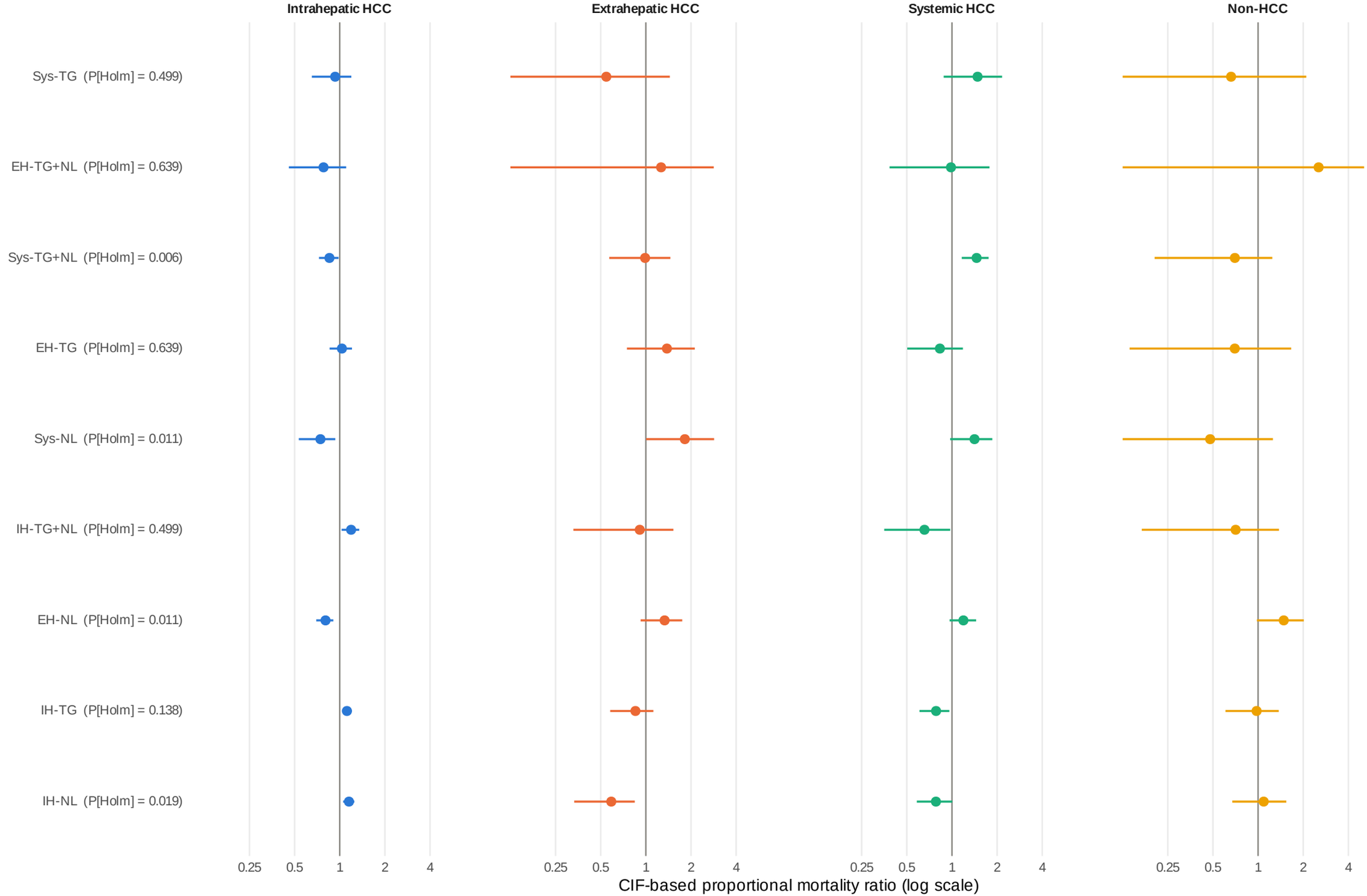
PMR = share of deaths from cause j within the pattern / share in all progressors. 95% bootstrap CI (clipped at 1/8 and 8).
P[Holm]: weighted-effect-coded Wald test (df = 2) that the pattern's csHR is equal across IH/EH/Systemic HCC death (from Lunn–McNeil model).



PMR is a proportion metric, not a risk metric: interpret alongside the absolute CIF matrix; PMRs within a pattern are mutually constrained.

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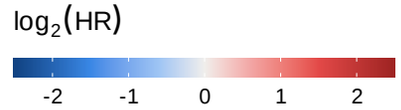


PMR is a proportion metric, not a risk metric: interpret alongside the absolute CIF matrix; PMRs within a pattern are mutually constrained.

Supplementary: Firth-penalised HR / csHR (reference: Sys-TG)

coxphf (Firth-penalised partial likelihood, Wald CI from penalised information); treatment group entered as covariate (coxphf has no strata)

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	0.57 (0.40–0.80)**	0.68 (0.43–1.10)	0.76 (0.23–2.49)	0.29 (0.16–0.52)***	NE (sparse reference)
IH-NL	0.49 (0.34–0.69)***	0.62 (0.38–1.01)	0.45 (0.13–1.53)	0.25 (0.13–0.46)***	NE (sparse reference)
IH-TG+NL	0.75 (0.51–1.10)	0.98 (0.59–1.65)	1.16 (0.31–4.39)	0.36 (0.17–0.76)**	NE (sparse reference)
EH-TG	0.88 (0.60–1.29)	0.96 (0.56–1.65)	2.01 (0.59–6.84)	0.51 (0.26–1.01)	NE (sparse reference)
EH-NL	0.61 (0.43–0.87)**	0.52 (0.32–0.86)*	1.37 (0.43–4.42)	0.48 (0.27–0.86)*	NE (sparse reference)
EH-TG+NL	0.86 (0.51–1.45)	0.76 (0.36–1.64)	2.18 (0.45–10.56)	0.59 (0.24–1.46)	NE (sparse reference)
Sys-TG	Reference	Reference	Reference	Reference	Reference (1 event)
Sys-NL	0.85 (0.57–1.27)	0.67 (0.37–1.20)	2.78 (0.80–9.64)	0.83 (0.43–1.59)	NE (sparse reference)
Sys-TG+NL	0.87 (0.61–1.24)	0.81 (0.49–1.33)	1.41 (0.42–4.68)	0.88 (0.50–1.56)	NE (sparse reference)
Overall Wald P (df=8)	**P < 0.001**	**P = 0.002**	**P < 0.001**	**P < 0.001**	**P = 0.303**

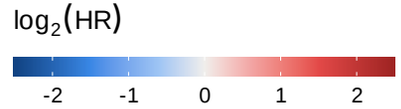


Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: HR / csHR excluding undetermined deaths (reference: Sys-TG)

n = 1778 (8 deaths with unclassified/unknown/missing cause removed)

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	0.58 (0.41–0.81)**	0.72 (0.45–1.17)	0.89 (0.26–3.08)	0.29 (0.16–0.52)***	NE (sparse reference)
IH-NL	0.50 (0.35–0.71)***	0.65 (0.40–1.07)	0.53 (0.15–1.91)	0.25 (0.13–0.46)***	NE (sparse reference)
IH-TG+NL	0.78 (0.53–1.15)	1.07 (0.64–1.81)	1.33 (0.33–5.31)	0.34 (0.16–0.73)**	NE (sparse reference)
EH-TG	0.89 (0.61–1.32)	1.02 (0.59–1.76)	2.35 (0.66–8.39)	0.49 (0.25–0.98)*	NE (sparse reference)
EH-NL	0.62 (0.44–0.88)**	0.54 (0.33–0.90)*	1.58 (0.47–5.35)	0.48 (0.27–0.85)*	NE (sparse reference)
EH-TG+NL	0.85 (0.50–1.43)	0.73 (0.34–1.60)	2.15 (0.41–11.34)	0.56 (0.22–1.43)	NE (sparse reference)
Sys-TG	Reference	Reference	Reference	Reference	Reference (1 event)
Sys-NL	0.84 (0.56–1.26)	0.67 (0.37–1.21)	3.08 (0.84–11.22)	0.81 (0.42–1.55)	NE (sparse reference)
Sys-TG+NL	0.88 (0.62–1.26)	0.82 (0.50–1.36)	1.69 (0.48–5.92)	0.89 (0.50–1.57)	NE (sparse reference)
Overall Wald P (df=8)	**P < 0.001**	**P = 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.419**



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: HR / csHR including patients flagged for deletion (reference: Sys-TG)

n = 1797

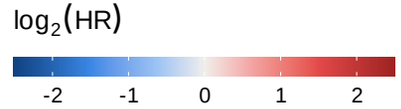


Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: HR / csHR with the earlier pattern rule (reference: Sys-TG)

STG + TTU and EHG + TTU classified as Sys-TG instead of Sys-TG+NL (22 patients reclassified)

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	0.54 (0.40–0.72)***	0.63 (0.43–0.92)*	0.75 (0.25–2.24)	0.28 (0.16–0.47)***	NE (sparse reference)
IH-NL	0.46 (0.34–0.62)***	0.57 (0.38–0.84)**	0.45 (0.14–1.40)	0.24 (0.14–0.42)***	NE (sparse reference)
IH-TG+NL	0.73 (0.52–1.02)	0.94 (0.61–1.46)	1.14 (0.32–3.99)	0.34 (0.16–0.68)**	NE (sparse reference)
EH-TG	0.84 (0.60–1.19)	0.90 (0.56–1.43)	2.01 (0.64–6.26)	0.48 (0.25–0.92)*	NE (sparse reference)
EH-NL	0.58 (0.43–0.78)***	0.48 (0.31–0.72)***	1.35 (0.46–3.95)	0.47 (0.28–0.78)**	NE (sparse reference)
EH-TG+NL	0.79 (0.48–1.29)	0.65 (0.31–1.33)	1.84 (0.39–8.67)	0.56 (0.23–1.36)	NE (sparse reference)
Sys-TG	Reference	Reference	Reference	Reference	Reference (1 event)
Sys-NL	0.78 (0.55–1.12)	0.59 (0.35–0.98)*	2.63 (0.83–8.33)	0.79 (0.43–1.45)	NE (sparse reference)
Sys-TG+NL	0.79 (0.58–1.07)	0.68 (0.44–1.03)	1.43 (0.47–4.35)	0.85 (0.50–1.44)	NE (sparse reference)
Overall Wald P (df=8)	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.397**



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.